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Big Data Analytics ICS 574 Final Project

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1. Introduction

This report details the machine learning project focused on analyzing credit card transaction data for fraud detection. The dataset was gotten from [Kaggle](#). It contains information of credit card transactions. This information includes transaction times, amounts, and associated personal and merchant details. According to the dataset author, the information is useful when conducting analysis such as fraud detection, customer segmentation, transaction classification, geospatial analysis, predictive modeling, behavioral analysis and anomaly detection. The primary objective was to build and evaluate predictive models capable of distinguishing fraudulent transactions from legitimate ones, enhancing the detection capabilities within financial systems.

2. Data Understanding and Exploration

2.1 Dataset Overview

The dataset used was derived from Kaggle and contains credit card transaction information. It includes 1,296,675 rows and 23 columns, covering transaction times, amounts, merchant details, and personal attributes of cardholders. The primary domain of this data is financial fraud detection, customer segmentation, transaction classification, and related predictive analytics.

2.2 Feature Description

Key features of the dataset include:

Table 1 Features Description

Column	Column Description	Data Type	Updated Data Type	Significance
Date_time	Timestamp of the transaction.	object	Categorical	Significant
cc_num	Credit card number	int64	Numerical	Not Significant
Merchant	Merchant or store where the transaction occurred.	object	Categorical	Significant
Category	Type of transaction (e.g., grocery, entertainment).	object	Categorical	Significant
Amt	Amount of the transaction.	float64	Numerical	Significant
First	First name of the cardholder.	object	Categorical	Not Significant
Last	Last name of the cardholder.	object	Categorical	Not Significant
Gender	Gender of the cardholder.	object	Categorical	Significant
Street	Address details of the cardholder.	object	Categorical	Not Significant
City	Address details of the cardholder.	object	Categorical	Significant
State	Address details of the cardholder.	object	Categorical	Not Significant

Zip	Address details of the cardholder.	int64	Numerical	Not Significant
Lat	Geographical coordinates of the transaction.	float64	Numerical	Significant
Long	Geographical coordinates of the transaction.	float64	Numerical	Significant
city_pop	Population of the city where the transaction occurred.	int64	Numerical	Significant
Job	Occupation of the cardholder.	object	Categorical	Significant
Dob	Date of birth of the cardholder.	object	Categorical	Significant
trans_num	Unique transaction number.	Object	Categorical	Not Significant
unix_time	Unix timestamp of the transaction.	int64	Numerical	Not Significant
merch_lat	Geographical coordinates of the merchant.	float64	Numerical	Not Significant
merch_long	Geographical coordinates of the merchant.	float64	Numerical	Not Significant
is_fraud	Indicator of whether the transaction is fraudulent	int64	Numerical	Significant
merch_zipcode	Geographical coordinates of the merchant.	float64	Numerical	Not Significant

2.3 Dataset Structure

The dataset consists of:

- **Number of rows:** 1,296,675
- **Number of columns:** 23

2.4 Missing Values and Duplicates

Missing data only exists in the merch_zipcode column and this missing data is about 15% of the total number values of the merch_zipcode column. There are no duplicate values(rows) in the dataset.

1 Exploring missing null values

```
#missing data
credit_data.isnull().sum()

trans_date_trans_time    0
cc_num                   0
merchant                 0
category                 0
amt                     0
first                   0
last                    0
gender                  0
street                  0
city                    0
state                   0
zip                     0
lat                     0
long                    0
city_pop                0
job                     0
dob                     0
trans_num               0
unix_time               0
merch_lat               0
merch_long              0
is_fraud                0
merch_zipcode           195973
dtype: int64

credit_data.isnull().mean() * 100

trans_date_trans_time    0
cc_num                   0
merchant                 0
category                 0
amt                     0
first                   0
last                    0
gender                  0
street                  0
city                    0
state                   0
zip                     0
lat                     0
long                    0
city_pop                0
job                     0
dob                     0
trans_num               0
unix_time               0
merch_lat               0
merch_long              0
is_fraud                0
merch_zipcode           15
dtype: float64
```

2.5 Statistical Summary

Initial exploration revealed that:

- **Amt (transaction amount)** and **city_pop (city population)** had high variance, with right-skewed distributions. The mean values were significantly higher than the medians.
- Summary statistics like mean, median, and standard deviation were computed to understand feature distribution.

Table 2 Statistical Summary

```
2]: credit_data.describe()
```

	cc_num	amt	zip	lat	long	city_pop	unix_time	merch_lat	merch_long	is_fraud	merch_zipcode
count	1296675	1296675	1296675	1296675	1296675	1296675	1296675	1296675	1296675	1296675	1100702
mean	417192042079726656	70	48801	39	-90	88824	1349243637	39	-90	0	46826
std	1308806447000789248	160	26893	5	14	301956	12841278	5	14	0	25834
min	60416207185	1	1257	20	-166	23	1325376018	19	-167	0	1001
25%	180042946491150	10	26237	35	-97	743	1338750742	35	-97	0	25114
50%	3521417320836166	48	48174	39	-87	2456	1349249747	39	-87	0	45860
75%	4642255475285942	83	72042	42	-80	20328	1359385376	42	-80	0	68319
max	4992346398065154048	28949	99783	67	-68	2906700	1371816817	68	-67	1	99403

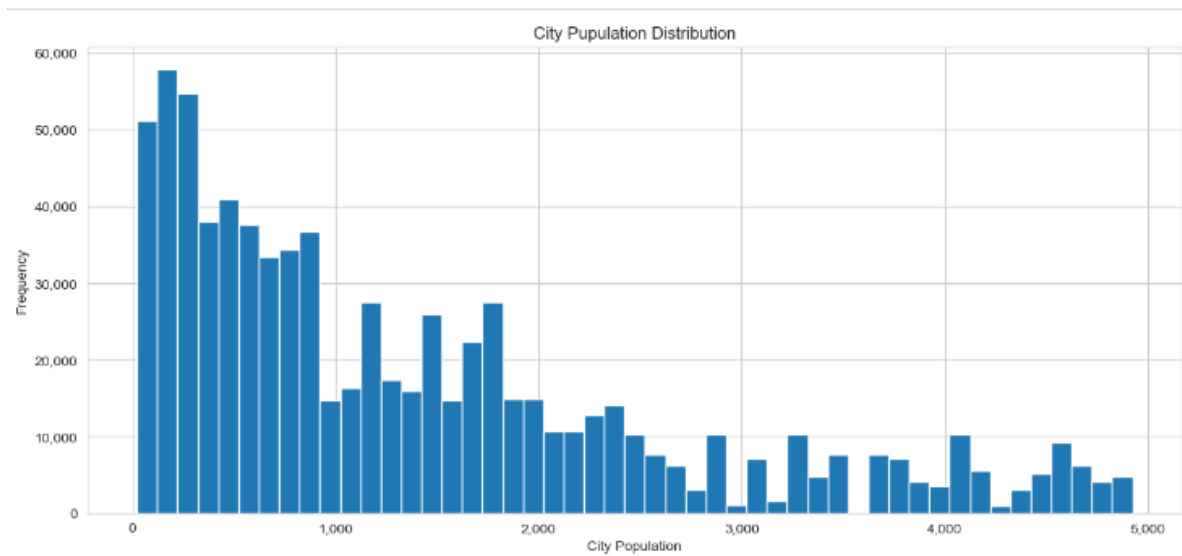
2.6 Data Distribution

Visual analyses indicated that:

- **Transaction amounts** were predominantly below \$20, with notable outliers.
- **City populations** were generally under 1,000, with outliers exceeding 5,000.
- Age distribution of cardholders showed a near-normal distribution, with most affected users aged 30 to 65.
- **Fraudulent transactions** displayed similar distribution patterns to non-fraudulent ones, albeit with some distinctions in specific categories.

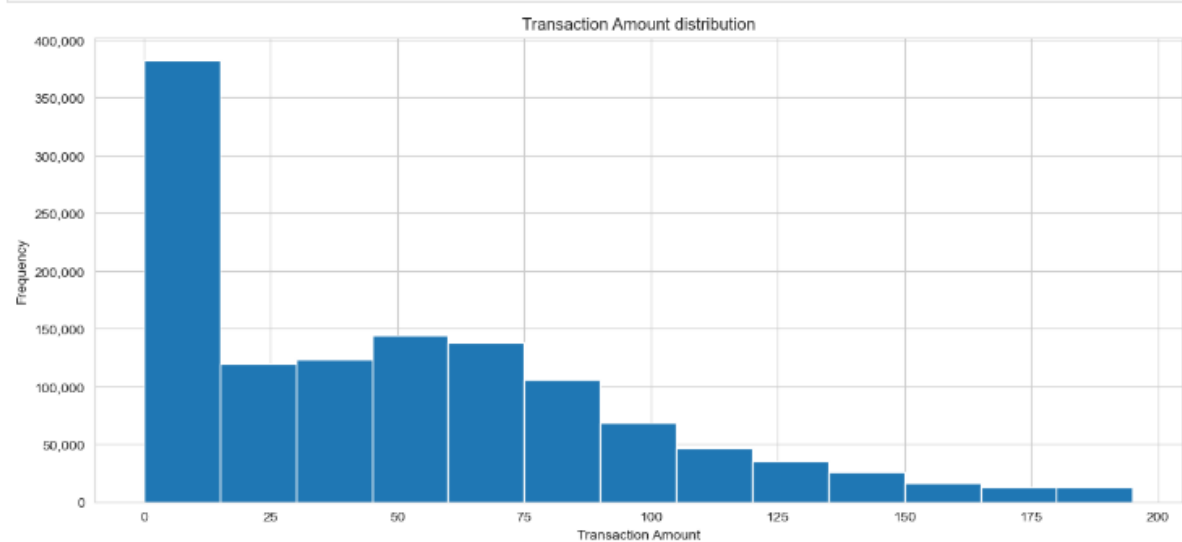
From the visualization above, both transaction amount and city population are right skewed with mean of each of the feature greater than the media. Furthermore, most of city population falls below 1000 and has outliers value above 5000. Additionally, most transactions fall below \$20 with some outliers above 200.

2 Distributions of cities population and amount spent in transactions



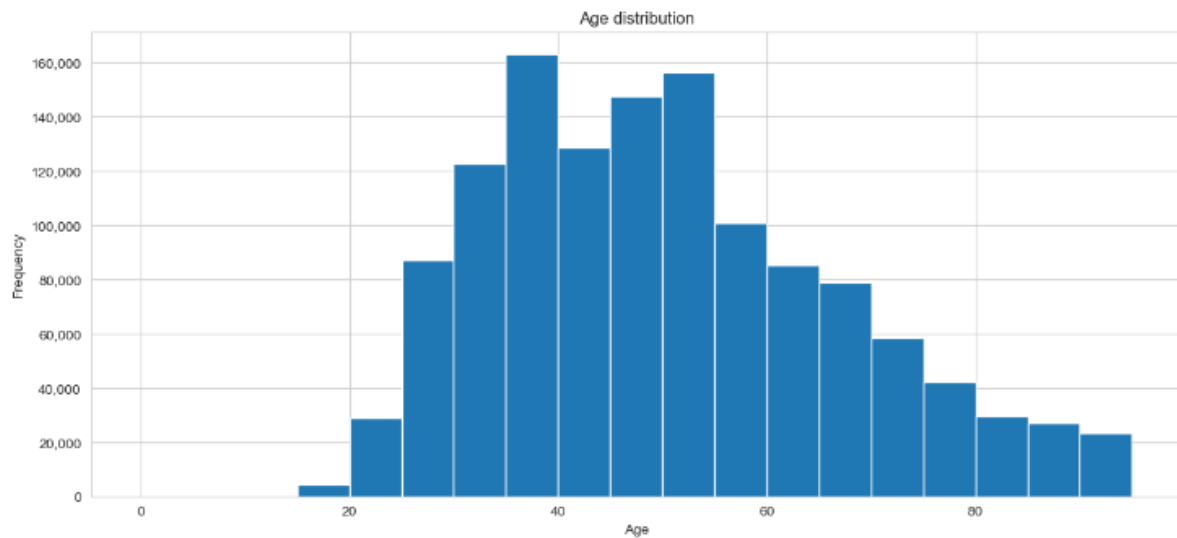
Transaction Amount Histogram

```
[130]: credit_data['amt'].hist(bins=np.arange(start=0, stop=200, step=15), figsize=[14,6])
plt.xlabel('Transaction Amount')
plt.ylabel('Frequency')
plt.title("Transaction Amount distribution")
plt.gca().xaxis.set_major_formatter(mticker.StrMethodFormatter('{x:,.0f}'))
plt.gca().yaxis.set_major_formatter(mticker.StrMethodFormatter('{x:,.0f}'))
```



Distribution of age of the data and fraudulent transaction data show similar trends. Both showed an almost normal distribution with much of the age of affected users falling between 30 to 65 years.

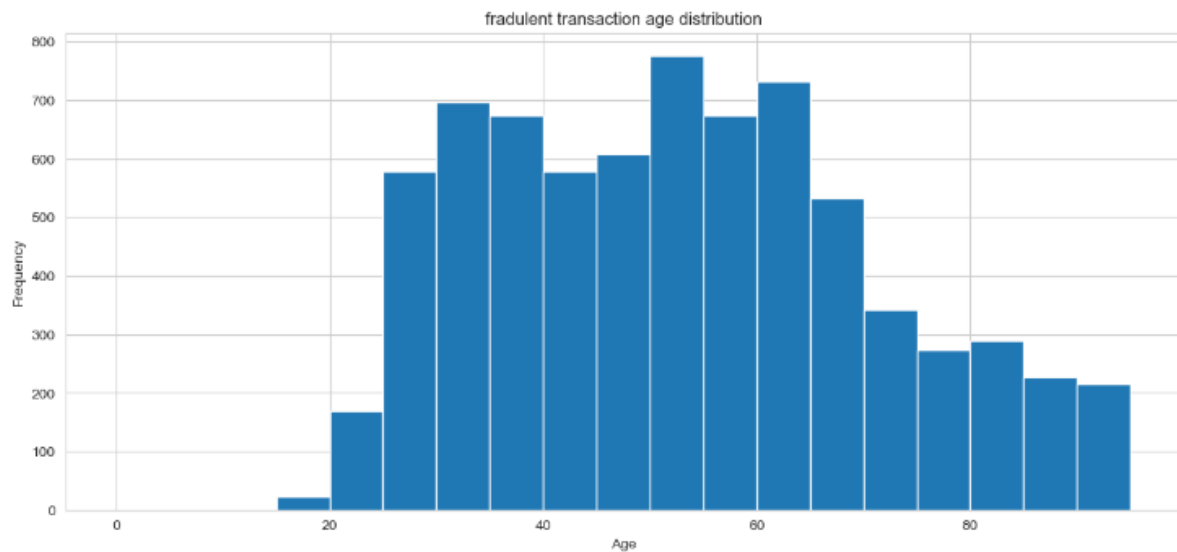
3 Distributions of Age and fraudulent transactions by age



```
[112]: min(credit_data['age'])
```

```
[112]: 19
```

```
[128]: fraud_transactions['age'].hist(bins=np.arange(start=0, stop=100, step=5), figsize=[14,6])
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.title("fraudulent transaction age distribution")
plt.gca().xaxis.set_major_formatter(mticker.StrMethodFormatter('{x:,.0f}'))
plt.gca().yaxis.set_major_formatter(mticker.StrMethodFormatter('{x:,.0f}'))
```



Correlation Analysis

4 Correlation Heatmap

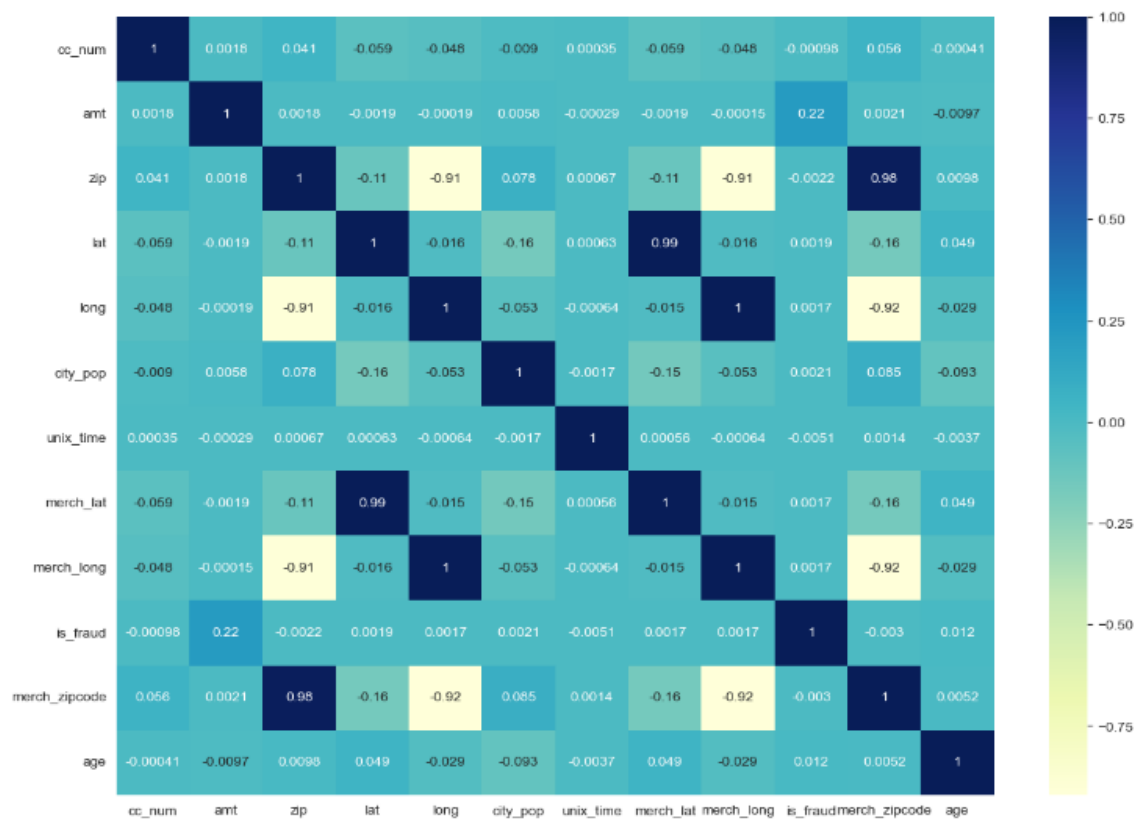
7. Correlation Analysis

What is the relationship between different features and the target variable?

- Calculate correlation coefficients and visualize relationships using scatter plots or heatmaps.

```
[133]: # Plotting correlation heatmap
plt.figure(figsize=(14, 10))
dataplot = sns.heatmap(credit_data.corr(numeric_only=True), cmap="YlGnBu", annot=True)

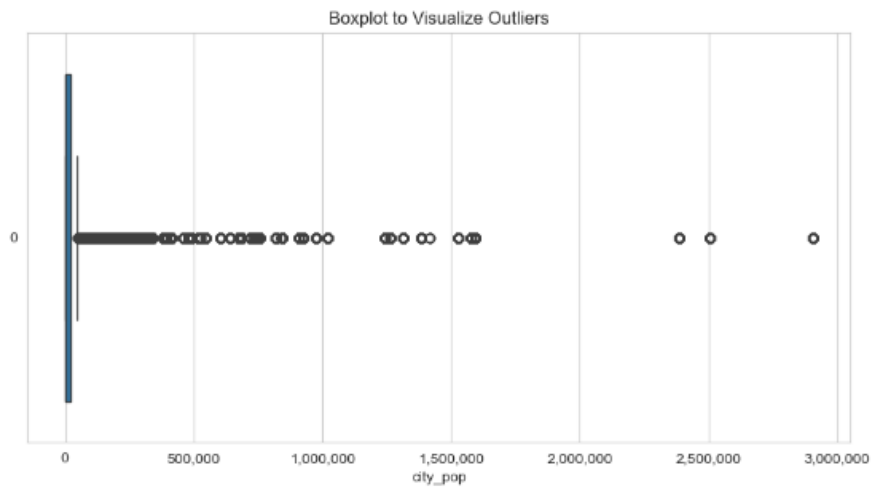
# Displaying heatmap
plt.show()
```



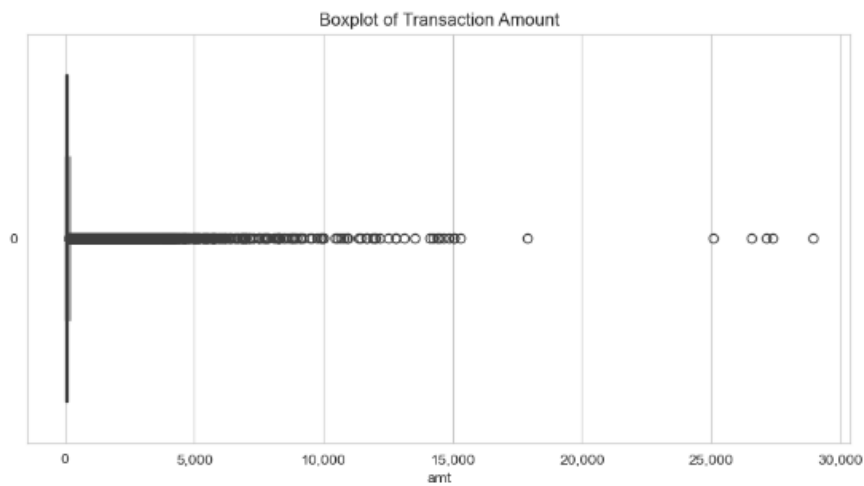
Outlier Detection

5 Boxplots to detect outliers for city population and amount spent in transactions

```
[143]: # Boxplot visualization
plt.figure(figsize=(10, 5))
sns.boxplot(x=credit_data["city_pop"])
plt.gca().xaxis.set_major_formatter(mticker.StrMethodFormatter('{x:,.0f}'))
plt.gca().yaxis.set_major_formatter(mticker.StrMethodFormatter('{x:,.0f}'))
plt.title('Boxplot of the Transaction City Population')
plt.show()
```

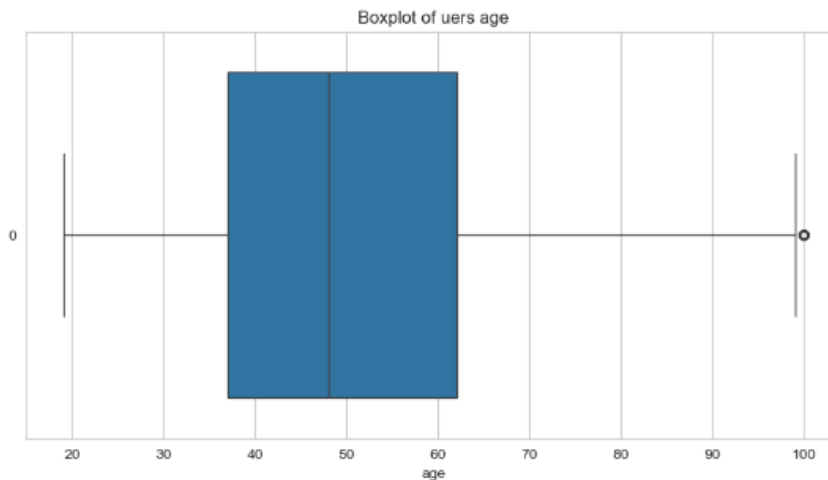


```
[148]: # Boxplot visualization
plt.figure(figsize=(10, 5))
sns.boxplot(x=credit_data["amt"])
plt.gca().xaxis.set_major_formatter(mticker.StrMethodFormatter('{x:,.0f}'))
plt.gca().yaxis.set_major_formatter(mticker.StrMethodFormatter('{x:,.0f}'))
plt.title('Boxplot of Transaction Amount')
plt.show()
```



6 Age Boxplot

```
[149]: # Boxplot visualization
plt.figure(figsize=(10, 5))
sns.boxplot(x=credit_data["age"])
plt.gca().xaxis.set_major_formatter(mticker.StrMethodFormatter('{x:,.0F}'))
plt.gca().yaxis.set_major_formatter(mticker.StrMethodFormatter('{x:,.0F}'))
plt.title('Boxplot of uers age')
plt.show()
```



3. Data Preprocessing

3.1 Handling Missing Data

Depending on the scale of the missing values, imputation, domain knowledge, prediction and deletion can be employed. In imputation, the mean, median and mode of the column can be input depending on the distribution of the column. If the missing values are significant, deletion can be employed. Domain knowledge can be used if there is background knowledge of the data. Furthermore, the analyst can predict the missing data.

In this project, missing data is only present in merch_zipcode column which is not significant in our analysis; hence, no action was taken to address missing data in this project

However, there's a number of transactions with credit card numbers that seem suspicious and invalid. This is most likely an entry error it might be attributed to the software not reading 0 if it was in the beginning of the card number. Based on this conclusion, we preprocessed the data to match 16 digits for regular cards (e.g. Visa) and 15 digits for MasterCard cards. We were able to identify the card types through some lines of codes displayed below along with the

7 Identifying card types based on credit card numbers and observing fraud frequencies per card type

```
In [11]: import re

def assign_card_type(cc_num):
    cc_num_str = str(cc_num)
    if re.match(r'^4\d{15}$', cc_num_str):
        return 'Visa'
    elif re.match(r'^5[1-5]\d{14}$', cc_num_str):
        return 'Mastercard'
    elif re.match(r'^3[47]\d{13}$', cc_num_str):
        return 'American Express'
    else:
        return 'Unknown'

df['card_type'] = df['cc_num'].apply(assign_card_type)

In [14]: print(df['card_type'].value_counts())
```

Unknown	941423
Visa	210830
American Express	123834
Mastercard	20588

Name: card_type, dtype: int64

```
In [18]: import pandas as pd

# Filter for fraudulent transactions
fraud_df = df[df['is_fraud'] == 1]

# Group the filtered DataFrame by card type and count the occurrences
fraud_count_by_card_type = fraud_df.groupby('card_type').size()

print(fraud_count_by_card_type)
```

card_type	
American Express	626
Mastercard	162
Unknown	5525
Visa	1193

dtype: int64

8 Preprocessing cards with unusual digit numbers

```
In [33]: # Check for unusual digit counts
df['cc_num_length'] = df['cc_num'].astype(str).str.len()
print(df['cc_num_length'].value_counts())

# Filter for cards with unusual lengths (e.g., less than 16 digits)
unusual_cards = df[df['cc_num_length'] < 16]
# print(unusual_cards)
```

16	649148
15	215968
14	125985
19	118789
12	91468
13	89206
11	6111

Name: cc_num_length, dtype: int64

```
In [20]: def identify_unusual_cards(row):
    if row['card_type'] == 'American Express':
        return row['cc_num_length'] < 15
    else:
        return row['cc_num_length'] < 16

df['unusual_length'] = df.apply(identify_unusual_cards, axis=1)

In [21]: def preprocess_card_number(cc_num, card_type):
    if card_type == 'Mastercard' and len(str(cc_num)) < 15:
        return str(cc_num).zfill(15)
    elif len(str(cc_num)) < 16:
        return str(cc_num).zfill(16)
    else:
        return cc_num

df['preprocessed_cc_num'] = df.apply(lambda row: preprocess_card_number(row['cc_num'], row['card_type']), axis=1)

In [29]: # Extract length during preprocessing
df['preprocessed_cc_num_length'] = df['preprocessed_cc_num'].str.len()
```

3.2 Encoding Categorical Variables

Yes, category columns like gender, age_group category were encoded. We employed a label encoding method. Value with the highest count in each category was given the highest label encoded number.

9 Categorical variables encoding 1

10. Encoding Categorical Variables

Are there categorical variables that need to be encoded?

- Describe the encoding techniques you will use (e.g., one-hot encoding, label encoding).

```
[127]: credit_data.columns

[127]: Index(['trans_date_trans_time', 'cc_num', 'merchant', 'category', 'amt',
        'first', 'last', 'gender', 'street', 'city', 'state', 'zip', 'lat',
        'long', 'city_pop', 'job', 'dob', 'trans_num', 'unix_time', 'merch_lat',
        'merch_long', 'is_fraud', 'merch_zipcode', 'age', 'age_group'],
        dtype='object')
```

Encoding column formula

```
[128]: # Encoding formula

def label_encode_by_frequency(df, column_name):
    # Count the frequency of each unique value in the specified column
    value_counts = df[column_name].value_counts()

    # Create a mapping of each value to a label based on frequency (highest count gets highest label)
    label_mapping = {value: rank for rank, value in enumerate(value_counts.index, start=1)}

    # Reverse the labels to ensure the highest frequency gets the highest number
    max_label = len(label_mapping)
    label_mapping = {key: max_label - label + 1 for key, label in label_mapping.items()}

    # Apply the label encoding to the specified column
    encoded_column_name = f"{column_name}_encoded"
    df[encoded_column_name] = df[column_name].map(label_mapping)

    return df, label_mapping
```

merchant column encoding

```
[129]: credit_data, mapping = label_encode_by_frequency(credit_data, 'merchant')
```

```
[ ]:
```

```
[ ]:
```

category column encoding

```
[130]: credit_data, mapping = label_encode_by_frequency(credit_data, 'category')
```

```
[ ]:
```

```
[131]: 'gender', 'city', 'state', 'job', 'age_group', 'merchant'
```

```
[131]: ('gender', 'city', 'state', 'job', 'age_group', 'merchant')
```

gender column encoding

```
[132]: credit_data, mapping = label_encode_by_frequency(credit_data, 'gender')
```

```
- -
```

10 Categorical variables encoding 2

```
[ ]:   
  
    city column encoding  
  
[133]: credit_data, mapping = label_encode_by_frequency(credit_data, 'city')  
  
[ ]:   
  
    state column encoding  
  
[134]: credit_data, mapping = label_encode_by_frequency(credit_data, 'state')  
  
    job column encoding  
  
[135]: credit_data, mapping = label_encode_by_frequency(credit_data, 'job')  
  
    age_group column encoding  
  
[136]: credit_data, mapping = label_encode_by_frequency(credit_data, 'age_group')  
  
[ ]:   
  
[137]: credit_data.columns  
  
[137]: Index(['trans_date_trans_time', 'cc_num', 'merchant', 'category', 'amt',  
         'first', 'last', 'gender', 'street', 'city', 'state', 'zip', 'lat',  
         'long', 'city_pop', 'job', 'dob', 'trans_num', 'unix_time', 'merch_lat',  
         'merch_long', 'is_fraud', 'merch_zipcode', 'age', 'age_group',  
         'merchant_encoded', 'category_encoded', 'gender_encoded',  
         'city_encoded', 'state_encoded', 'job_encoded', 'age_group_encoded'],  
         dtype='object')
```

3.3 Feature Scaling

Features like **Amt** and **city_pop** required scaling due to their skewed distributions. Standard scaling was applied to normalize these features for model readiness.

3.4 Feature Selection

Features were selected based on their correlation with the target variable, `is_fraud`. Key features included **Amt**, **category_encoded**, **city_pop**, and **gender_encoded**.

4. Model Building and Evaluation

4.1 Model Selection

Three machine learning models were trained and evaluated:

- **XGBoost Classifier**
- **Random Forest Classifier**
- **Logistic Regression**

The **XGBoost Classifier** demonstrated the highest accuracy and predictive power.

▼ Which features will you include in your model, and why?

- Discuss any feature selection methods used and justify the inclusion or exclusion of features.

```
[146]: #Defining base Learners

myclf1 = KNeighborsClassifier(n_neighbors=5)
myclf2 = DecisionTreeClassifier(max_depth=5)
myclf3 = GaussianNB()
myclf4 = RandomForestClassifier(n_estimators=100, random_state=1)
myclf5 = LinearSVC(random_state=1)
myclf6 = LogisticRegression(random_state=1)
myclf7 = XGBClassifier()

[147]: # @title Updated Stacking Model
# Define base classifiers with names
estimators = [
    ("KNeighborsClassifier", myclf1),
    ("DecisionTreeClassifier", myclf2),
    ("GaussianNB", myclf3),
    ("RandomForestClassifier", myclf4),
    ("LinearSVC", myclf5),
    ("LogisticRegression", myclf6),
    ("XGBClassifier", myclf7)
]

# Initialize StackingClassifier with the List of named estimators
stackingclf = StackingClassifier(estimators=estimators, final_estimator=LogisticRegression())
```

Model Training With Scaler Transformed Data

```
[148]: # @title Model Training With Scaler Transformed Data
# Separate the target variable (assuming it's named 'target')
X = credit_data.drop(['is_fraud'], axis=1)
y = credit_data['is_fraud']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20, random_state = 0)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

[149]: # @title Training models & checking accuracy
# Train and evaluate individual classifiers
classifiers = {
    "KNeighborsClassifier": myclf1,
    "DecisionTreeClassifier": myclf2,
    "GaussianNB": myclf3,
    "RandomForestClassifier": myclf4,
    "LinearSVC": myclf5,
    "LogisticRegression": myclf6,
    "XGBClassifier": myclf7
}

for name, clf in classifiers.items():
    clf.fit(X_train, y_train)
    y_pred = clf.predict(X_test)
    accuracy = accuracy_score(y_test, y_pred)
    print(f"{name} Accuracy: {accuracy}")

KNeighborsClassifier Accuracy: 0.996147839668402
DecisionTreeClassifier Accuracy: 0.9962866562554226
GaussianNB Accuracy: 0.9911620105269247
RandomForestClassifier Accuracy: 0.99748587734012
LinearSVC Accuracy: 0.9940424547400081
LogisticRegression Accuracy: 0.9935990128598146
XGBClassifier Accuracy: 0.9972159561956543
```


4.2 Feature Importance

The most impactful features were:

- **Amt**
- **category_encoded**
- **gender_encoded**
- **city_encoded**

These were determined using permutation importance and built-in feature importance from the models.

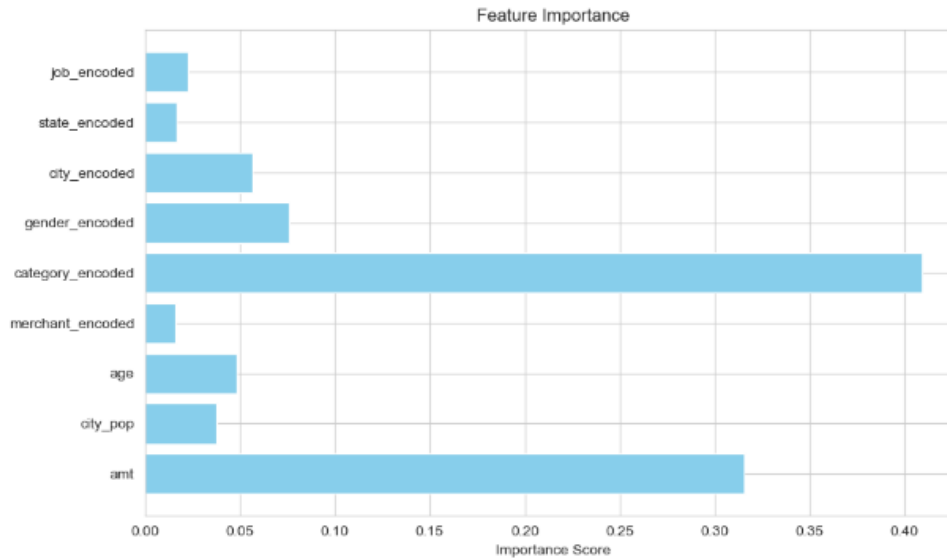
12 Important features for XGBoost Classifier

• After training your model, visualize feature importance using bar charts (e.g., for tree-based models) or coefficients (e.g., for linear models).

```
[150]: #Important features for XGBoost Classifier

# Extract feature importance
importances = myclf7.feature_importances_
feature_names = X.columns

# Visualize the feature importance
plt.figure(figsize=(18, 6))
plt.barh(feature_names, importances, color='skyblue')
plt.xlabel('Importance Score')
plt.title('Feature Importance')
plt.show()
```



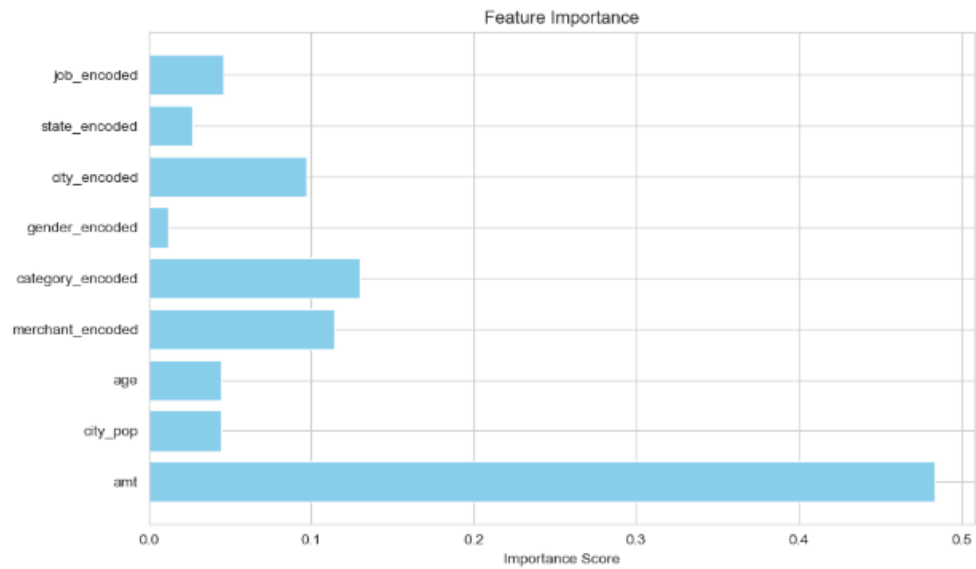
13 Important Features for RandomForestClassifier

Importance Score

```
[151]: #important features for RandomForestClassifier
```

```
# Extract feature importance
importances = myclf4.feature_importances_
feature_names = X.columns

# Visualize the feature importance
plt.figure(figsize=(10, 6))
plt.barh(feature_names, importances, color='skyblue')
plt.xlabel('Importance Score')
plt.title('Feature Importance')
plt.show()
```

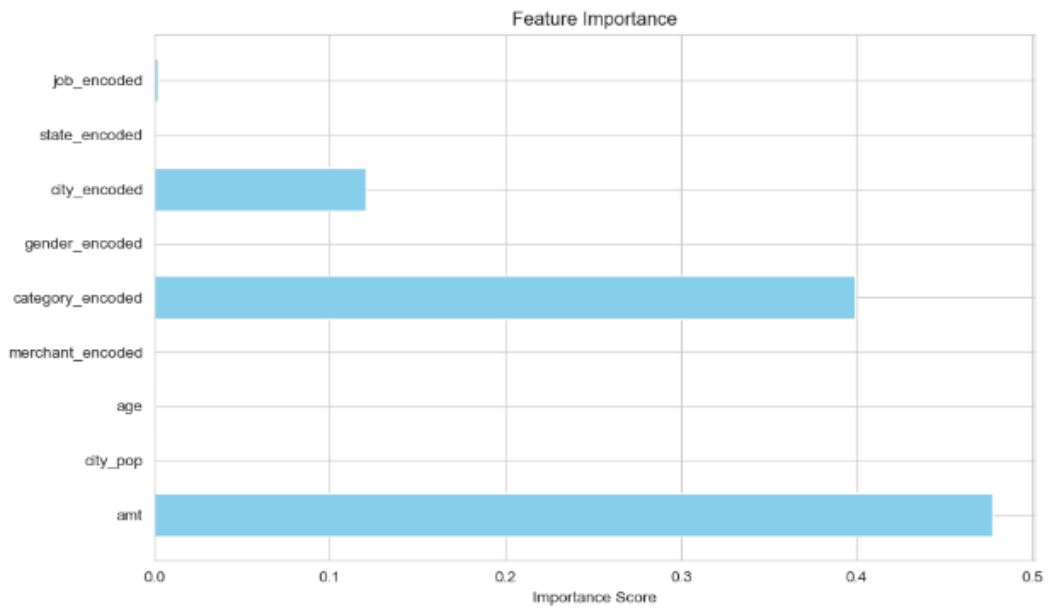


14 Important Features for DecisionTreeClassifier

```
#Important features for DecisionTreeClassifier

# Extract feature importance
importances = myclf2.feature_importances_
feature_names = X.columns

# Visualize the feature importance
plt.figure(figsize=(10, 6))
plt.barh(feature_names, importances, color='skyblue')
plt.xlabel('Importance Score')
plt.title('Feature Importance')
plt.show()
```

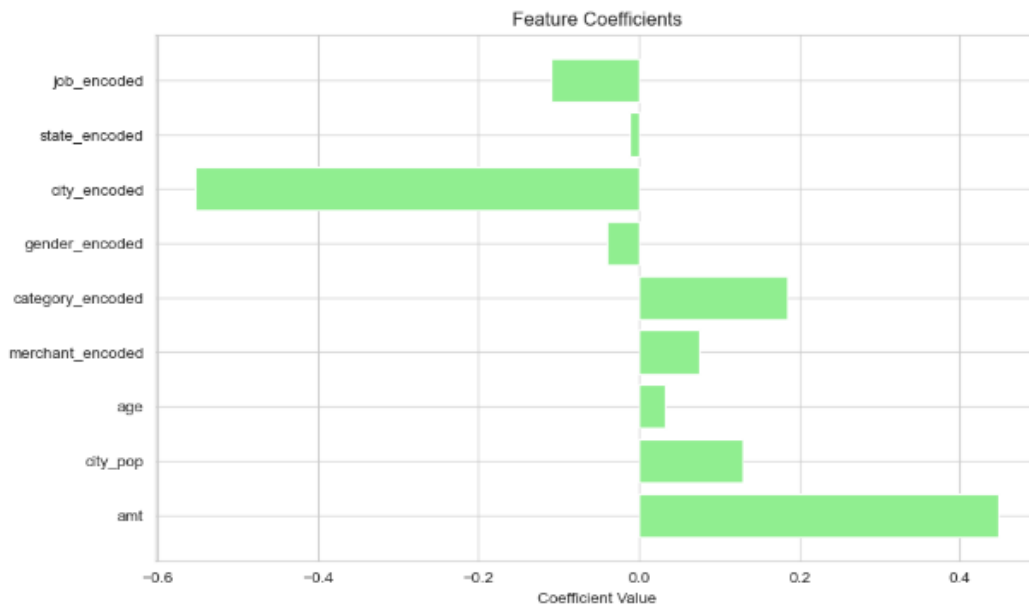


15 Coefficients for Logistic Regression

```
# coefficient for LogisticRegression

coefficients = myclf6.coef_[0]
feature_names = X.columns

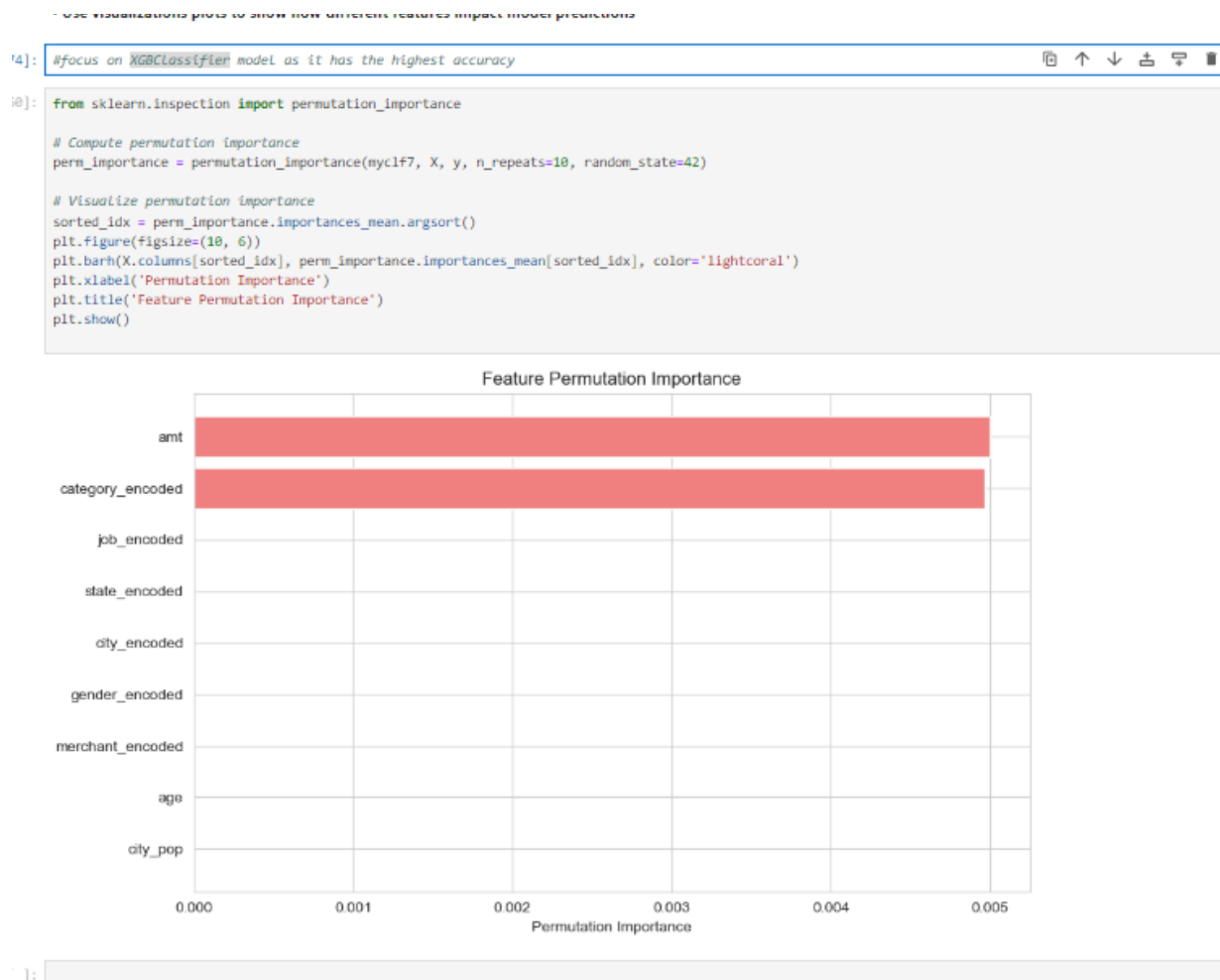
# Visualize the coefficients
plt.figure(figsize=(10, 6))
plt.barh(feature_names, coefficients, color='lightgreen')
plt.xlabel('Coefficient Value')
plt.title('Feature Coefficients')
plt.show()
```



4.3 Model Performance

The XGBoost Classifier outperformed other models with the highest accuracy and balanced precision-recall metrics. Visualizations indicated that **Amt** and **Category** significantly influenced the predictions.

16 XGBoost Feature Permutation Importance



5. Visual Analysis

5.1 Fraud Distribution by City Population

Cities with high populations showed varied fraud incidences, with visual mapping highlighting high-risk areas.

5.2 Category and Merchant Analysis

Certain transaction categories, such as **entertainment** and **high-value purchases**, showed increased fraud rates.

5.3 Age Group Susceptibility

Age segmentation analysis revealed that middle-aged individuals (30-65) were more frequently associated with fraud, aligning with the general distribution of the dataset.

6. Conclusion

The project successfully leveraged machine learning to identify patterns in credit card transactions related to fraud. The **XGBoost Classifier** was identified as the most effective model, with key features including **Amt** and **Category**. Future work could focus on integrating additional real-time behavioral data for even more robust fraud detection.

7. Recommendations

- Incorporate real-time transaction monitoring.
- Enhance model robustness with new, high-resolution data.
- Implement continuous model training for evolving fraud patterns.

Thank you for reading & Kind Regards

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